INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

Department of Civil Engineering

PROJECT REPORT

Deep Learning-Based Ocean Wind Field Estimation Using Sentinel-1 SAR Imagery Over Indian Coastal Waters

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Github Repository - https://github.com/rajaditya-code/CEC_OPEN_PROJECT

ABSTRACT

Ocean surface wind is one of the most consequential environmental parameters governing weather prediction, ocean circulation, offshore renewable energy assessment, and maritime navigation. Conventional observational networks comprising moored buoys, meteorological stations, and research vessels offer reliable point measurements but provide insufficient spatial coverage over the vast expanses of the open ocean. Satellite remote sensing, particularly through Synthetic Aperture Radar (SAR), offers a powerful complement by enabling all-weather, day-and-night, high-resolution observations of ocean surface roughness, which is directly modulated by near-surface wind conditions.

This study presents a complete deep learning framework for estimating ocean wind speed and direction from Sentinel-1 C-band SAR imagery over selected coastal regions of India, specifically the Tamil Nadu and Gujarat coastlines. A modified ResNet-34 convolutional neural network was trained on 515 dual-polarized SAR image patches, each of dimensions 256 x 256 pixels with VV and VH polarization channels, paired with corresponding ERA5 atmospheric reanalysis wind labels. A temporally separated dataset was employed, with 2024 observations used for training and validation, and independent 2023 observations constituting the final test set of 309 samples.

To address the circular nature of wind direction, sine-cosine encoding was applied to directional targets, enabling stable regression learning. The model was trained using transfer learning from ImageNet, a progressive backbone unfreezing strategy, cosine learning-rate scheduling, and Automatic Mixed Precision (AMP) computation on an NVIDIA Tesla T4 GPU. During validation, the model achieved a wind speed Mean Absolute Error (MAE) of 3.328 m/s and a wind direction MAE of 70.03 degrees. On the independent test dataset, the model achieved a wind speed MAE of 4.394 m/s and RMSE of 4.521 m/s. Wind direction estimation proved more challenging, with a test MAE of 133.44degrees, attributed to the coarse resolution of ERA5 reference data, the limited training dataset size, and inherent ambiguity in SAR backscatter patterns.

A FastAPI-based deployment framework was implemented to provide automated wind field predictions and vector visualisations for user-defined coastal areas of interest.

The study demonstrates the feasibility of end-to-end deep learning for operational SAR-based ocean wind estimation over Indian coastal waters, and establishes a foundation for future improvements in dataset scale, model architecture, and label quality.

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CHAPTER 1: INTRODUCTION

1.1 Background and Context

Ocean surface wind is among the most consequential environmental parameters in Earth science, governing the exchange of momentum, heat, and moisture between the atmosphere and the ocean. Wind-driven processes underpin thermohaline circulation, coastal upwelling, tropical cyclone intensification, and the formation of surface wave fields. Accurate, spatially comprehensive knowledge of ocean surface wind conditions is therefore indispensable for operational meteorology, climate modelling, offshore energy planning, marine transportation, fisheries management, and coastal disaster preparedness.

Conventional wind observations are gathered through a sparse network of meteorological stations, research vessels, moored buoys, and offshore platforms. Although reliable, these in-situ measurements offer only limited spatial coverage across the vast oceanic regions of the world. Numerical weather prediction (NWP) models attempt to fill observational gaps by assimilating available measurements, but they remain computationally demanding and may inadequately represent fine-scale wind variability in complex coastal environments characterised by steep orography, land-sea thermal contrasts, and shallow water bathymetry.

Satellite remote sensing has emerged as a transformative complement to conventional approaches, enabling synoptic-scale and globally consistent monitoring of ocean surface conditions. Among the available satellite technologies, Synthetic Aperture Radar (SAR) is uniquely suited to ocean wind estimation. SAR systems transmit microwave pulses and record the backscattered energy from the Earth's surface. The intensity of this backscatter over the ocean is strongly modulated by wind-driven capillary waves and short gravity waves, which roughen the sea surface in proportion to local wind conditions. Because SAR operates at microwave frequencies, it can acquire high-resolution imagery under all weather conditions and at any time of day, overcoming the cloud-cover limitations of optical sensors.

The Sentinel-1 mission, developed by the European Space Agency (ESA) under the Copernicus Earth Observation Programme, provides freely accessible C-band SAR imagery with global coverage and a revisit frequency of six to twelve days. With its Interferometric Wide Swath (IW) imaging mode delivering 250 km swath coverage at approximately ten metres spatial resolution, Sentinel-1 has become the reference SAR dataset for a wide range of oceanographic applications. The open distribution policy of Copernicus has enabled the remote sensing research community to develop and validate diverse applications, including sea ice mapping, oil spill detection, ship detection, and ocean wind retrieval.

Recent advances in deep learning have substantially expanded the capability to extract quantitative environmental information from satellite imagery. Convolutional Neural Networks (CNNs) learn hierarchical feature representations directly from raw image data without requiring explicit feature engineering, enabling robust performance across diverse image analysis tasks. Residual Networks (ResNets), introduced by He et al. (2016), employ skip connections to facilitate gradient flow through deep architectures, enabling efficient training and strong generalisation even with moderate-size training datasets. Their success across remote sensing tasks, including land cover classification, change detection, and parameter retrieval, motivates their application to ocean wind estimation from SAR imagery.

This project presents a complete deep learning framework for estimating ocean wind field parameters — specifically wind speed and wind direction — from Sentinel-1 SAR imagery over selected coastal regions of India. The framework integrates satellite data acquisition through Google Earth Engine, preprocessing and ocean masking, ERA5 reanalysis label generation, model training using a modified ResNet-18 architecture, and operational deployment through a FastAPI web service. The study focuses on the Tamil Nadu and Gujarat coastlines, which represent two of India's most important offshore wind resource zones.

1.2 Problem Statement

India possesses a coastline exceeding 7,500 km, spanning the Arabian Sea, the Bay of Bengal, and the Indian Ocean. Accurate and timely ocean wind information in these waters is essential for offshore renewable energy characterisation, maritime navigation and safety, monsoon monitoring, coastal disaster management, and fisheries resource assessment. Current in-situ observational infrastructure — comprising a limited number of moored buoys, coastal meteorological stations, and irregular ship-of-opportunity measurements — provides only sparse and temporally discontinuous coverage over the surrounding ocean areas.

NWP models and global reanalysis products such as ERA5 offer broader spatial coverage, but typically operate at horizontal resolutions of 25-31 km. This spatial resolution is insufficient to resolve the fine-scale wind variability that governs coastal upwelling, sea breeze dynamics, and the performance of individual offshore wind turbines. Satellite SAR, with its metre-scale resolution, can in principle provide wind estimates at spatial scales inaccessible to NWP systems. However, translating SAR backscatter into accurate wind speed and direction estimates is a non-trivial inverse problem.

Conventional SAR-based wind retrieval relies on empirical geophysical model functions (GMFs) such as the CMOD family. While these models have been extensively validated for open-ocean conditions, they require external wind direction information from NWP models, which can introduce errors in complex coastal environments. Furthermore, CMOD models are based on fixed analytical parameterisations that may not fully capture the nonlinear and scene-dependent relationships between SAR backscatter and wind conditions. A data-driven deep learning approach that learns these relationships directly from matched SAR and wind observation pairs offers a potentially more flexible and accurate alternative.

1.3 Motivation

India's energy transition strategy identifies offshore wind power as a strategic priority. The Ministry of New and Renewable Energy has established ambitious targets for offshore wind capacity, with the Arabian Sea coast of Gujarat and the Bay of Bengal coast of Tamil Nadu identified as the most promising development zones owing to their consistently high wind resources and established maritime infrastructure. Accurate wind resource characterisation at high spatial resolution — the spatial detail that SAR can provide — is a prerequisite for effective wind farm siting, turbine specification, and long-term energy yield estimation.

Beyond energy, the Indian Ocean basin is one of the world's most cyclone-active regions. The Bay of Bengal, which borders Tamil Nadu, experiences frequent intense tropical cyclones during pre-monsoon and post-monsoon seasons. High-resolution ocean wind monitoring capability would contribute directly to improved early warning systems and disaster management operations. The increasing availability of free satellite data, cloud-based geospatial platforms such as Google Earth Engine, and accessible deep learning frameworks creates an unprecedented opportunity to develop sophisticated operational monitoring systems at minimal cost.

1.4 Objectives

The primary objective of this project is to develop and evaluate a deep learning-based framework for estimating ocean wind speed and wind direction from Sentinel-1 SAR imagery over selected Indian coastal regions. The specific objectives are as follows:

1. Acquire Sentinel-1 dual-polarized (VV and VH) SAR imagery over the Tamil Nadu and Gujarat coastal regions through Google Earth Engine.
2. Collect temporally co-located ERA5 atmospheric reanalysis wind data as reference labels for model training and evaluation.
3. Generate ocean-only image patches using water masking techniques based on the JRC Global Surface Water dataset.
4. Develop a modified ResNet-18 convolutional neural network for simultaneous estimation of wind speed and wind direction.
5. Implement sine-cosine encoding for wind direction to handle the inherent circularity of directional variables.
6. Train the model using a temporally separated dataset to enable realistic performance evaluation free from information leakage.
7. Evaluate model performance using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) for both wind speed and direction.
8. Deploy the trained model through a FastAPI-based web service providing automated wind field prediction and visualisation.

1.5 Scope and Limitations

The project focuses exclusively on ocean wind field estimation over selected coastal regions of India, using Sentinel-1 GRD imagery in Interferometric Wide Swath (IW) mode and ERA5 reanalysis data as reference. The scope includes data acquisition, preprocessing, model development, training, evaluation, and API-based deployment. The study does not extend to cyclone prediction, wave forecasting, ocean current estimation, or long-term climate trend analysis. Model generalisation outside the Tamil Nadu and Gujarat study regions is not evaluated. Furthermore, the absence of in-situ buoy measurements means that the model is trained and evaluated against ERA5 reanalysis data, which itself carries uncertainties arising from its 31 km spatial resolution.

1.6 Report Organisation

This report is organised into seven chapters. Chapter 2 reviews relevant literature on SAR-based wind retrieval, deep learning techniques, and the datasets employed. Chapter 3 describes the study regions, input datasets, and data acquisition and preparation methodology. Chapter 4 presents the system architecture, model design, and training strategy. Chapter 5 details the implementation environment and deployment framework. Chapter 6 presents and discusses the experimental results. Chapter 7 summarises the principal findings, acknowledges the limitations of the study, and outlines directions for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Ocean Wind Retrieval from Satellite Platforms

Satellite-based ocean wind retrieval has a history spanning more than four decades. Scatterometers were among the first spaceborne instruments deployed for this purpose, exploiting the sensitivity of radar backscatter to ocean surface roughness to retrieve both wind speed and direction from multiple azimuth angles. Missions such as NASA QuikSCAT and EUMETSAT ASCAT (operating on the Metop series of satellites) have demonstrated sustained, global wind measurement capability, and their data products are routinely assimilated into NWP systems. However, scatterometers typically offer spatial resolutions of 12.5 to 50 km, limiting their utility for coastal and nearshore applications. Passive microwave radiometers, such as the Special Sensor Microwave Imager (SSM/I), provide wind speed estimates at comparable resolutions but lack directional information.

Synthetic Aperture Radar provides the highest spatial resolution among satellite wind sensors, making it particularly valuable for coastal environments. Monaldo et al. (2001) conducted foundational comparisons of SAR-derived wind speeds against buoy observations and model predictions, establishing that SAR-based retrievals could achieve accuracy levels comparable to scatterometers at resolutions several orders of magnitude finer. Subsequent research has explored the use of SAR for offshore wind resource assessment (Hasager et al., 2011), wind ramp event detection, and tropical cyclone wind field mapping. Koch (2004) made important contributions to the understanding of SAR wind direction retrieval, demonstrating that streaks in SAR images associated with atmospheric roll vortices can be used to estimate wind direction through spectral analysis.

2.2 Sentinel-1 and Its Oceanographic Applications

The Sentinel-1 mission, initiated with the launch of Sentinel-1A in April 2014 and subsequently supplemented by Sentinel-1B (launched 2016), provides C-band SAR imagery under the European Union's Copernicus programme (Torres et al., 2012). The mission was designed specifically to ensure operational continuity and open data access, making it the most widely used SAR mission in the research community. The Interferometric Wide Swath (IW) mode, the primary acquisition mode over land and ocean, provides 250 km swath width with a ground resolution of approximately 5 metres in range and 20 metres in azimuth for ground range detected (GRD) products, which are multilooked to approximately 10 metres.

Sentinel-1 dual-polarization data — specifically the VV and VH polarization combinations — offer complementary oceanographic information. The VV channel, in which both the transmitted and received signals are vertically polarised, is most directly sensitive to Bragg resonance from surface waves and is the primary channel used in CMOD-based wind retrieval. The VH cross-polarization channel, which is less sensitive to the incident angle and is dominated by volume scattering at high wind speeds, has been shown to extend the dynamic range of wind speed retrieval to extreme conditions (Mouche et al., 2017) that saturate VV-based estimates. Using both channels as complementary inputs to a deep learning model therefore offers a more information-rich representation than either channel alone.

Table 2.1: Key characteristics of the Sentinel-1 SAR mission.

Parameter	Specification
Frequency Band	C-band (5.405 GHz)
Imaging Mode	Interferometric Wide Swath (IW)
Polarisation	VV + VH (dual-pol)
Spatial Resolution (GRD)	~10 m
Swath Width	250 km
Revisit Time	6 days (two satellites)
Data Access	Free via Copernicus Open Access Hub

2.3 Conventional Geophysical Model Functions

Before the advent of machine learning approaches, SAR ocean wind retrieval was dominated by empirical geophysical model functions (GMFs). The CMOD family is the most widely adopted GMF for C-band SAR. CMOD4, developed by Stoffelen and Anderson (1997), parameterises the normalised radar cross-section (NRCS) as a function of wind speed, radar incidence angle, and the relative angle between the wind direction and the radar look direction. Subsequent versions refined the parameterisation: CMOD5 improved accuracy at moderate wind speeds, CMOD5.N extended its validity to near-neutral atmospheric stability conditions, and CMOD7 was calibrated using modern Sentinel-1 data, improving agreement with collocated scatterometer retrievals.

Despite their widespread adoption, the CMOD models have inherent limitations. They require an external estimate of wind direction, typically sourced from a collocated NWP model, which introduces an additional source of uncertainty — particularly in coastal regions where NWP models may not accurately represent local wind patterns. Furthermore, CMOD models are physically motivated but empirically parameterised, meaning they may not generalise well to conditions outside the range of their training data or to polarisation channels (such as VH) not included in their original development. These limitations motivate the development of data-driven approaches capable of learning complex SAR-wind relationships without requiring explicit physical parameterisation.

2.4 Machine Learning Approaches

Machine learning methods were introduced to SAR wind retrieval as alternatives to GMFs, offering the ability to model nonlinear input-output relationships without requiring explicit physical equations. Li and Lehner (2014) applied support vector regression to wind speed retrieval from high-resolution TerraSAR-X images, demonstrating improved accuracy at fine spatial scales. Random Forest and gradient boosting methods have also been investigated for SAR wind applications, with results generally competitive with or superior to CMOD-based retrievals in specific conditions. However, classical machine learning methods require handcrafted feature extraction from SAR images — such as texture metrics, backscatter moments, or spectral features — which may not fully exploit the spatial structure present in high-resolution imagery.

Deep Convolutional Neural Networks (CNNs) overcome this limitation by learning features directly from raw imagery through an end-to-end training process. Zhang et al. (2019) demonstrated that CNN-based models trained on SAR imagery could retrieve ocean wind speeds with accuracies comparable to or exceeding CMOD-based methods, without requiring external wind direction inputs. The ability of CNNs to simultaneously exploit both local texture features and broader spatial context makes them particularly well-suited to problems where the relationship between image content and the target parameter is spatially heterogeneous and nonlinear.

2.5 Deep Learning in Remote Sensing

Deep learning has transformed remote sensing image analysis over the past decade. CNNs have achieved state-of-the-art performance across a broad range of remote sensing tasks, including land cover classification, change detection, building and road extraction, flood mapping, cloud detection, and quantitative parameter retrieval. The availability of large pretrained models — particularly those trained on the ImageNet dataset, which contains over one million labelled natural images — has enabled effective transfer learning for remote sensing applications where large labelled datasets are not available. Transfer learning exploits the generic low-level and mid-level features learned from natural images (edges, textures, shapes) that are also informative in SAR and optical remote sensing contexts.

Residual Networks (ResNets), introduced by He et al. (2016), represented a breakthrough in deep architecture design by demonstrating that very deep networks (50-152 layers) could be trained effectively using shortcut skip connections. The insight underlying ResNets is that it is easier to learn a residual mapping $F(x)$ than a direct mapping $H(x)$, where x is the input and the network learns $F(x) = H(x) - x$. The original ResNet paper demonstrated that residual networks with 18-152 layers outperformed previous architectures on ImageNet classification. ResNet-18, the shallowest variant, offers a balance between feature extraction capability and computational efficiency, making it well-suited for regression tasks on relatively small datasets.

2.6 Wind Direction Estimation Challenges

While wind speed estimation from SAR imagery has reached a reasonable level of maturity, accurate wind direction retrieval remains substantially more challenging. There are several fundamental reasons for this asymmetry. First, ocean surface roughness — the primary physical quantity governing SAR backscatter — is more directly related to wind speed than to wind direction. The backscatter intensity increases monotonically with wind speed (within the dynamic range of the instrument) but varies only weakly with direction, making direction a less constrained quantity given the observed backscatter. Second, similar SAR backscatter signatures can be produced by winds arriving from different directions, particularly when atmospheric roll vortices and other mesoscale atmospheric structures are absent.

From a machine learning perspective, wind direction presents an additional challenge arising from its circular nature: 0 degrees and 360 degrees represent the same physical direction, yet they are numerically distant if treated as a real-valued regression target. Naive regression of wind direction angles produces large errors near the wrap-around point. The standard solution is to represent direction using its sine and cosine components, which provides a continuous and differentiable representation across the full 360-degree range. During inference, the predicted sine and cosine values are converted back to angle using the four-quadrant arctangent function. This approach was adopted in the present study and has been demonstrated to substantially improve directional prediction stability in the deep learning literature.

2.7 ERA5 Reanalysis Data

ERA5 is the fifth-generation global atmospheric reanalysis product produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) (Hersbach et al., 2020). It provides hourly global fields of atmospheric variables at approximately 31 km horizontal resolution and 137 vertical levels, spanning from 1940 to the present. ERA5 assimilates observations from satellites, radiosondes, surface stations, aircraft, and ships using a state-of-the-art four-dimensional variational (4D-Var) data assimilation scheme. Among the variables provided, the 10 m eastward (U10) and northward (V10) wind components are the standard reference for ocean wind applications. These are combined to compute wind speed as the magnitude of the horizontal wind vector and wind direction as the meteorological direction using the four-quadrant arctangent function.

While ERA5 represents the current state of the art in global reanalysis, it is important to recognise its limitations as a label source for SAR wind estimation. The spatial resolution of 31 km is substantially coarser than the 10 m resolution of Sentinel-1 GRD imagery, meaning that a single ERA5 pixel covers an area equivalent to approximately 9.4 million Sentinel-1 pixels. Fine-scale wind variability present in SAR imagery – associated with atmospheric boundary layer rolls, coastal jets, and land-sea breezes — is not resolved by ERA5. This resolution mismatch introduces noise into the label assignment process, which is one of the primary factors contributing to prediction uncertainty, particularly for wind direction.

2.8 Research Gap Analysis and Contributions

Despite significant progress in SAR-based wind retrieval, a number of important research gaps remain in the existing literature. Most published studies have focused on European coastal waters, the North Atlantic, and the North Sea, where dense networks of offshore platforms and meteorological buoys provide validation opportunities. Comparable studies focusing specifically on Indian coastal environments — where the monsoon system, tropical cyclone activity, and distinct coastal geometry present unique challenges — are largely absent. Furthermore, many published works emphasise wind speed estimation and give limited attention to the co-estimation of wind direction from the same SAR image. End-to-end frameworks that integrate automated data acquisition, deep learning inference, and operational deployment through a web API are also largely absent from the published literature.

The present study addresses these gaps by: (1) focusing specifically on Indian coastal waters covering two distinct oceanographic regimes (the Arabian Sea and the Bay of Bengal); (2) developing a unified model for simultaneous wind speed and direction estimation using dual-polarization Sentinel-1 imagery; (3) adopting a rigorous temporal train-test separation to prevent information leakage and provide realistic performance estimates; and (4) implementing a FastAPI-based deployment framework that transforms the research prototype into an operational prediction tool accessible through a standard web interface.

CHAPTER 3: STUDY AREA AND DATA DESCRIPTION

3.1 Study Regions

The study focuses on two distinct coastal regions of India selected for their oceanographic importance, offshore wind energy potential, and availability of Sentinel-1 observations. The Tamil Nadu coast, situated along the southeastern margin of the Indian subcontinent, borders the Bay of Bengal and the Gulf of Mannar. The Gujarat coast, on the western margin, fronts the Arabian Sea and the Gulf of Khambhat. Together, these two regions sample markedly different atmospheric and oceanographic regimes, providing the dataset with a degree of climatological diversity that improves model generalisation.

3.1.1 Tamil Nadu Coastal Region

Tamil Nadu experiences a strongly seasonal climate controlled by the Southwest (June-September) and Northeast (October-December) monsoons. During the Southwest monsoon, southerly and southwesterly winds drive coastal upwelling along the Tamil Nadu shelf, while the Northeast monsoon brings northeasterly winds and associated wave activity that significantly influences coastal morphology and fisheries productivity. The coastal waters near Chennai, Nagapattinam, Cuddalore, and Tuticorin were included in the study, covering a latitudinal range from approximately 8.5 deg N to 13.5 deg N and a longitudinal range from 79.5 deg E to 80.7 deg E.

3.1.2 Gujarat Coastal Region

Gujarat possesses the longest coastline of any Indian state, extending approximately 1,600 km along the Arabian Sea. The region is characterised by strong seasonal winds, with the Southwest monsoon bringing powerful southwesterly flow across the Arabian Sea during summer months. Gujarat has been identified by the Indian government as the primary zone for offshore wind energy development, with significant installed capacity planned in the Gulf of Khambhat and surrounding waters. The study covered a latitudinal range of approximately 20 deg N to 24 deg N and a longitudinal range of 68 deg E to 72 deg E.

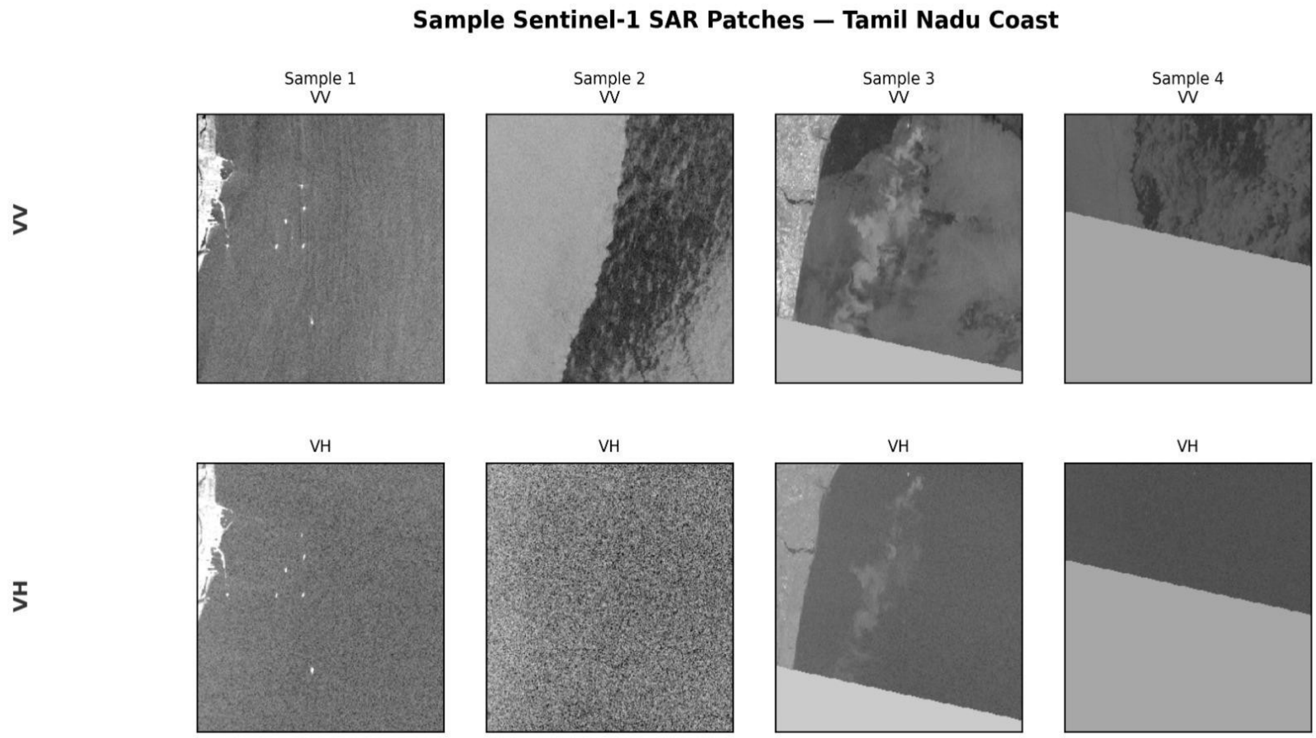
Table 3.1: Geographical extent of the study regions.

Region	Latitude Range	Longitude Range
Tamil Nadu (Bay of Bengal)	8.5 deg N - 13.5 deg N	79.5 deg E - 80.7 deg E
Gujarat (Arabian Sea)	20.0 deg N - 24.0 deg N	68.0 deg E - 72.0 deg E

3.2 Sentinel-1 SAR Dataset

Sentinel-1 Ground Range Detected (GRD) products acquired in Interferometric Wide Swath (IW) mode were used as the primary data source. GRD products are multilooked and map-projected, providing amplitude images at approximately 10 m pixel spacing suitable for backscatter analysis. The dual-polarization configuration combining VV and VH channels was selected, as both channels are available in the standard IW-mode acquisition over ocean. Imagery was accessed through the Google Earth Engine platform, which hosts the complete Sentinel-1 GRD archive and enables efficient cloud-based querying, filtering, and preprocessing.

For each study region, Sentinel-1 scenes were filtered by acquisition mode (IW), product type (GRD), and polarisation availability (VV and VH). Multiple scenes covering the same region on the same acquisition date were mosaicked to produce spatially continuous coverage. The resulting mosaics were then processed to extract normalised backscatter coefficients in decibels for both polarisation channels.



Picture presents representative Sentinel-1 GRD SAR patches extracted from the Tamil Nadu coastal region. Both VV and VH polarization channels were used as model inputs. The dual-polarization information provides complementary surface scattering characteristics that are useful for wind estimation.

3.3 ERA5 Atmospheric Reanalysis

ERA5 hourly wind data were retrieved from the Copernicus Climate Data Store for dates corresponding to Sentinel-1 acquisitions. The 10 m eastward wind component (U10) and northward wind component (V10) were extracted and spatially matched to the centre coordinates of each SAR image patch. Wind speed was computed as WS = sqrt(U10^2 + V10^2), expressed in metres per second. Wind direction was computed using the meteorological convention as theta = arctan2(U10, V10) converted to degrees in the range 0 to 360 degrees. These wind speed and direction values served as the supervised learning targets for model training.

3.4 JRC Global Surface Water Dataset

The JRC Global Surface Water dataset, produced by Pekel et al. (2016) from a long time series of Landsat imagery, provides per-pixel estimates of the fraction of time each surface location has been classified as water. The water occurrence layer, which records the percentage of Landsat observations classified as water, was used to generate binary ocean masks for each candidate SAR image patch. A patch was retained for training or testing only if more than 85 percent of its pixels had a water occurrence value exceeding 50 percent, ensuring that the dataset was free from significant land contamination. This masking procedure removed patches covering coastal intertidal zones, rivers, estuaries, and land areas that would introduce irrelevant backscatter features into the training data.

3.5 Data Acquisition and Dataset Generation

Data acquisition and initial processing were performed entirely within the Google Earth Engine cloud computing environment, which provides direct access to the Sentinel-1 and JRC datasets without the need to download and locally store large volumes of raw satellite data. Python scripts using the Google Earth Engine API were developed to automate the full acquisition pipeline: querying and filtering the Sentinel-1 archive, mosaicking scenes, applying the JRC water mask, extracting fixed-size image patches, retrieving temporally matched ERA5 wind data, and assembling the final dataset tensors.

Each image patch was extracted at a fixed size of 256 x 256 pixels, corresponding to an approximate ground coverage of 2.56 km x 2.56 km. The two polarization channels (VV and VH) were stacked to form a two-channel input tensor of shape (2, 256, 256). Each channel was independently normalised using the z-score transformation (subtracting the channel mean and dividing by the standard deviation), ensuring that both channels contributed comparably to the learning process regardless of their absolute backscatter magnitude. The final dataset consisted of 515 training samples and 309 test samples.

Study area = Tamil nadu coast ***SAR SOURCE= Sentinel-1 GRD***

Table 3.2: Summary of the training and test datasets.

Property	Training Dataset	Test Dataset
Number of Samples	515	309
Input Channels	2 (VV, VH)	2 (VV, VH)
Patch Size (pixels)	256 x 256	256 x 256
SAR Tensor Shape	(515, 2, 256, 256)	(309, 2, 256, 256)
Label Tensor Shape	(515, 2)	(309, 2)
Label Variables	Wind Speed, Wind Direction	Wind Speed, Wind Direction

3.6 Temporal Train-Test Split

A deliberate temporal split strategy was adopted to ensure that the test dataset is genuinely independent of the training data. Random train-test splitting is commonly used in machine learning benchmarks but can introduce information leakage in meteorological applications, because similar atmospheric patterns may recur on short timescales and appear in both training and test subsets. This leakage leads to overly optimistic performance estimates that do not reflect real-world generalisation capability.

To avoid this problem, Sentinel-1 imagery acquired during five months of 2024 (January, March, May, August, and October) was used for model training and validation, while imagery from three months in the preceding year, 2023 (January, June, and October), was reserved exclusively for final evaluation. This temporal separation ensures that the model has no access to 2023 data during training and that its test performance reflects true temporal generalisation across different seasonal conditions. Training on 2024 data and testing on 2023 data is an unusual but methodologically sound design that prevents data leakage while exposing the model to different seasonal wind conditions in both splits.

CHAPTER 4: METHODOLOGY AND SYSTEM DESIGN

4.1 System Architecture Overview

The proposed framework is an end-to-end system that transforms raw Sentinel-1 SAR imagery into operational ocean wind field estimates. The system comprises six sequential processing stages: (1) satellite data acquisition through Google Earth Engine; (2) preprocessing, normalisation, and ocean masking; (3) training dataset construction with ERA5 wind labels; (4) deep learning model training and validation; (5) inference and wind vector reconstruction; and (6) operational deployment via a FastAPI web service. Each stage is automated through Python scripts, enabling the system to process new user-defined regions of interest with minimal manual intervention.

4.2 Data Preprocessing and Ocean Masking

Raw Sentinel-1 GRD backscatter values were extracted in units of decibels (dB) for both VV and VH polarisation channels. Each channel was normalised independently using the z-score transformation to ensure numerical stability during neural network training. The normalisation statistics (mean and standard deviation) were computed from the training dataset and applied consistently to the test dataset and during inference, preventing any information from the test or inference data from influencing the normalisation parameters.

Ocean masking using the JRC Global Surface Water dataset was applied to ensure that all training and test patches represented open-ocean or coastal water conditions. The 85 percent water fraction threshold was chosen to allow inclusion of patches in coastal transition zones while excluding patches where land scattering effects would dominate the backscatter signal. Data augmentation was applied during training to increase the effective diversity of the training set. Specifically, random horizontal flipping and the addition of low-amplitude Gaussian noise were applied. These augmentations help reduce overfitting and improve robustness to minor variations in image content.

4.3 Label Generation and Encoding

Wind speed labels were computed as the Euclidean magnitude of the ERA5 horizontal wind vector: WS = sqrt(U10^2 + V10^2), where U10 and V10 are the eastward and northward 10 m wind components. Wind direction labels were computed as the meteorological wind direction using the four-quadrant arctangent function and converted to degrees in the range 0 to 360 degrees.

Because wind direction is a circular variable — 0 degrees and 360 degrees being physically identical — direct regression of wind direction angles leads to large prediction errors near the wrap-around point. To address this, the direction angle theta was encoded as two Cartesian components: Y_sin = sin(theta) and Y_cos = cos(theta). The model therefore predicts a three-element output vector: [wind speed, sin(direction), cos(direction)]. During inference, the direction angle is recovered as theta = arctan2(Y_sin, Y_cos), and the result is mapped to the 0-360 degree range. This encoding ensures that the target variables vary continuously as a function of direction, removing the discontinuity that would otherwise impair model training.

4.4 Deep Learning Architecture

4.4.1 Selection of ResNet-34

ResNet-34 was selected as the backbone architecture based on several practical considerations and dual channel SAR input and spite regression heads. Its moderate parameter count (approximately 11 million parameters for standard RGB input) provides a good balance between feature extraction capability and training efficiency, which is important given the limited training set of 515 samples. Pretrained ImageNet weights are publicly available for ResNet34, enabling transfer learning that compensates for the small dataset size. The residual skip connections in ResNet-34 facilitate stable gradient flow during fine-tuning, reducing the risk of vanishing gradient problems that can impair training with small learning rates.

4.4.2 Network Modification for SAR Input

The standard ResNet-34 architecture is designed for three-channel RGB images. Since Sentinel-1 SAR data contains two polarisation channels (VV and VH), the first convolutional layer was modified to accept two-channel input tensors. The original layer has dimensions (64, 3, 7, 7) — 64 output filters with 3 input channels and a 7x7 kernel. In the modified architecture, this layer is replaced with a (64, 2, 7, 7) layer. The pretrained weights for the first layer were adapted by averaging the weights across the three original input channels and assigning the average to each of the two new input channels, a standard technique for adapting pretrained networks to non-standard input dimensionalities.

4.4.3 Output Layer Design

The fully connected classification head of the original ResNet-34 (designed for 1,000 ImageNet classes) was replaced with a custom regression head consisting of a linear layer mapping the 512-dimensional feature vector produced by the ResNet backbone to a 3-element output vector: [wind speed (m/s), sin(direction), cos(direction)]. No activation function was applied to the output layer, enabling unconstrained regression. The loss function was Mean Squared Error (MSE) computed across all three output dimensions simultaneously, with equal weighting applied to each component.

Table 4.1: Model training hyperparameter configuration.

Parameter	Value
Architecture	Modified ResNet-34
Input Channels	2 (VV and VH polarisation)
Output Dimensions	3 (speed, sin(dir), cos(dir))
Pretrained Weights	ImageNet
Optimizer	AdamW
Loss Function	Mean Squared Error (MSE)
Initial Learning Rate	0.001
Fine-Tuning Learning Rate	0.0003
Weight Decay	1 x 10^-4
Batch Size	16
Total Epochs	70
Warm-Up / Frozen Epochs	10 (Phase 1)
Fine-Tuning Epochs	60 (Phase 2)
LR Schedule	Cosine Annealing
Mixed Precision	Enabled (PyTorch AMP)
Training GPU	NVIDIA Tesla T4

4.5 Training Strategy

4.5.1 Transfer Learning

Training deep networks from scratch on small datasets is prone to overfitting and typically requires extensive regularisation. Transfer learning addresses this by initialising the network with weights pretrained on a large, diverse dataset — in this case, the ImageNet dataset comprising 1.2 million labelled natural images across 1,000 categories. Although SAR images are physically distinct from natural photographs, the low-level and mid-level feature representations learned from natural images (oriented edges, textures, blobs, and patterns) are broadly useful for processing any image data. Transfer learning from ImageNet has been demonstrated to consistently improve performance and reduce training time in remote sensing applications where labelled data is limited.

4.5.2 Progressive Backbone Unfreezing

To stabilise training and prevent catastrophic forgetting of the pretrained features, a two-phase training strategy was adopted. In Phase 1 (Epochs 1-10), the ResNet backbone was frozen and only the custom regression head was trained. This allowed the output layer to adapt its weights to the SAR wind prediction task without disrupting the pretrained feature representations. The learning rate during Phase 1 was set to 0.001. In Phase 2 (Epochs 11-70), the backbone was unfrozen and the entire network was fine-tuned end-to-end with a reduced learning rate of 0.0003, applied in conjunction with a cosine annealing schedule. This graduated approach prevents the large gradient updates that can occur when an untrained head forces large weight updates through a pretrained backbone during early training.

4.5.3 Cosine Learning Rate Scheduling

Following backbone unfreezing at Epoch 11, a cosine annealing learning rate schedule was applied for the remaining 60 epochs. Cosine annealing smoothly reduces the learning rate according to a cosine function from the initial fine-tuning rate (0.0003) to a minimum value approaching zero. This schedule avoids the abrupt discontinuities associated with step decay schedules and has been empirically demonstrated to improve final model performance by facilitating convergence to better local optima during the later stages of training.

4.5.4 Automatic Mixed Precision Training

Automatic Mixed Precision (AMP) training, implemented through the PyTorch torch.cuda.amp interface, was enabled throughout the training process. AMP dynamically selects between 16-bit (half-precision) and 32-bit (full-precision) floating point representations for different computational operations, reducing GPU memory consumption and increasing arithmetic throughput on hardware that supports Tensor Core operations. The NVIDIA Tesla T4 GPU used in this study supports FP16 Tensor Core operations, enabling significant acceleration of the training process without any loss of model accuracy.

4.5.5 Architecture Figure :

1. Sentinel-1 SAR

↓

2. Ocean Masking

↓

3. Patch Extraction

↓

4 .ERA5 Label Generation

↓

5. ResNet-34

↓

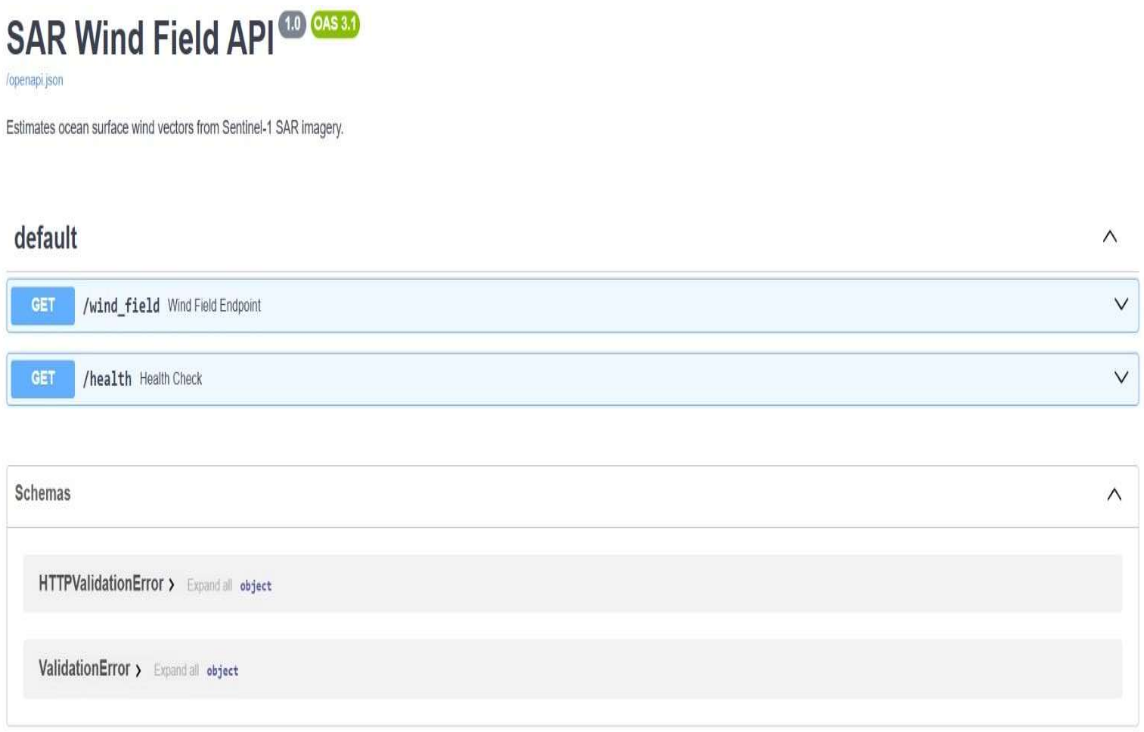
6. Speed + Sinθ + Cosθ

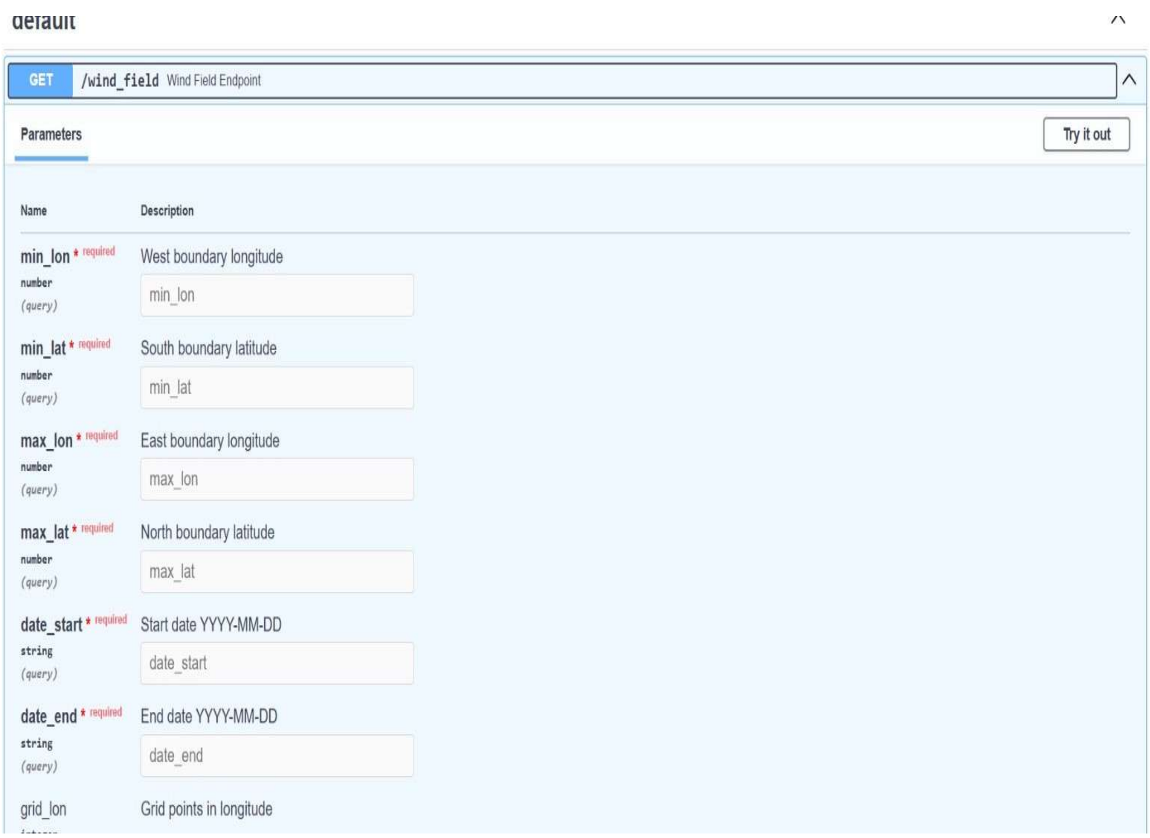
↓

7. Wind Vector Reconstruction

↓

8.FastAPI Deployment


SAR Wind Field API
1.0
OAS 3.1
/openapi.json
Estimates ocean surface wind vectors from Sentinel-1 SAR imagery.
default
GET /wind_field Wind Field Endpoint
GET /health Health Check
Schemas
HTTPValidationError Expand all object
ValidationError Expand all object


default
GET /wind_field Wind Field Endpoint
Parameters
Try it out
Name
Description
min_lon * required
number
(query)
West boundary longitude
min_lon
min_lat * required
number
(query)
South boundary latitude
min_lat
max_lon * required
number
(query)
East boundary longitude
max_lon
max_lat * required
number
(query)
North boundary latitude
max_lat
date_start * required
string
(query)
Start date YYYY-MM-DD
date_start
date_end * required
string
(query)
End date YYYY-MM-DD
date_end
grid_lon
Grid points in longitude

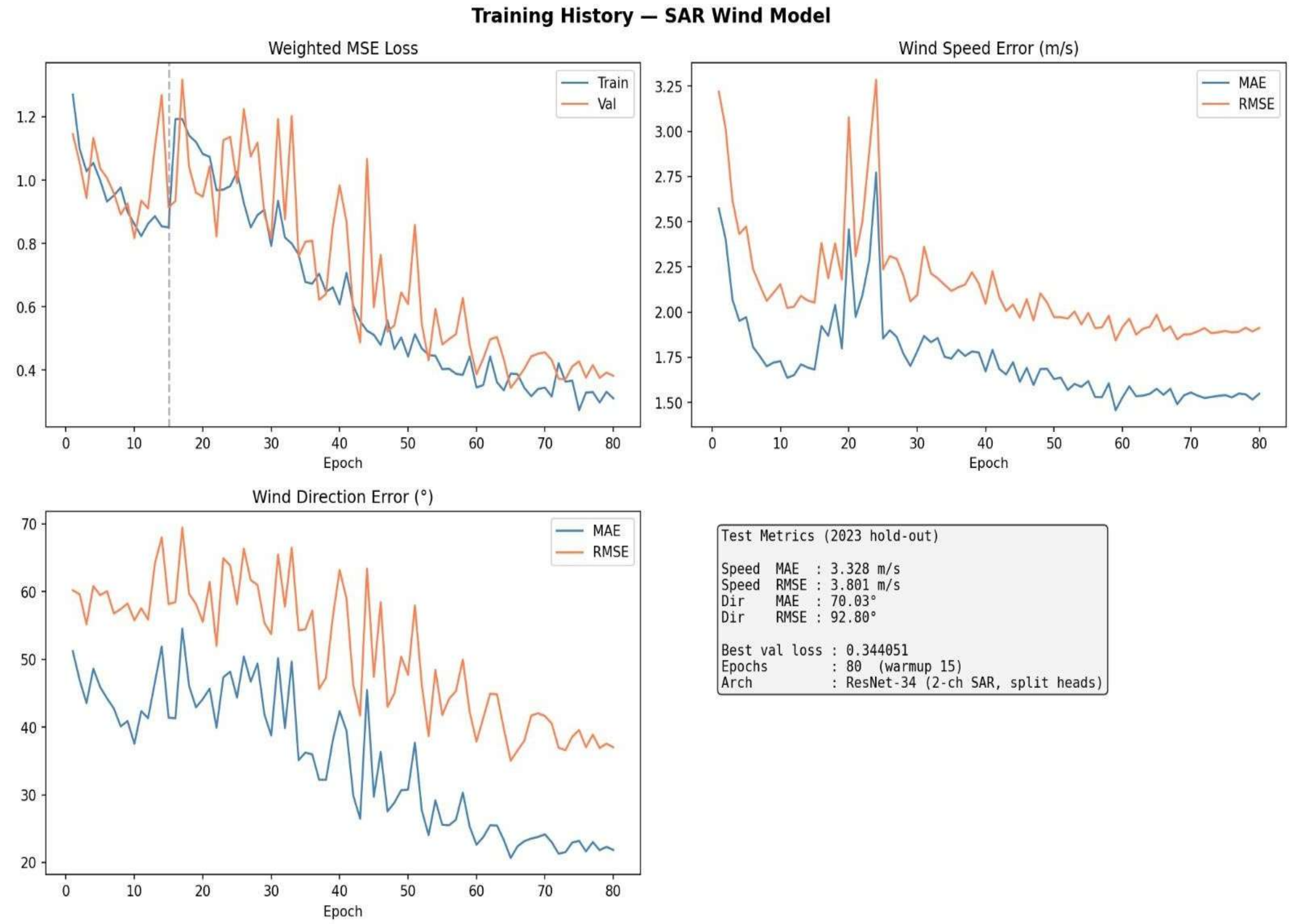
CHAPTER 5: IMPLEMENTATION AND DEPLOYMENT

5.1 Development Environment

The complete system was implemented using Python 3.x and the PyTorch deep learning framework. Model training was performed on an NVIDIA Tesla T4 GPU provided through the Google Colab cloud computing platform. The Tesla T4 is equipped with 16 GB of GDDR6 memory and supports FP16 Tensor Core operations, making it well-suited for mixed-precision deep learning workloads. Data storage and inter-session persistence were managed through Google Drive.

Table 5.1: Hardware and software configuration.

Component / Library	Specification / Purpose
GPU	NVIDIA Tesla T4 (16 GB)
RAM	16 GB
Platform	Google Colab (cloud-based)
Python	3.x – core programming language
PyTorch	Deep learning framework and training
Torchvision	Pretrained ResNet-18 weights
NumPy / Pandas	Numerical computation and data handling
Google Earth Engine API	Satellite data acquisition
Rasterio	Raster data processing
Albumentations	Data augmentation pipeline
FastAPI / Uvicorn	REST API development and serving
Matplotlib	Wind field visualisation
Scikit-Learn	Performance metric computation



5.2 Dataset Preparation Pipeline

The dataset preparation pipeline was automated through a series of Python scripts using the Google Earth Engine API. For each target month, the Sentinel-1 GRD collection was queried and filtered by acquisition mode, polarisation availability, and geographic region. Qualifying scenes were mosaicked to produce spatially continuous coverage of each study area. Image patches of 256 x 256 pixels were then extracted at regular spatial intervals across each mosaic, with a minimum spatial separation of one patch width between adjacent extraction points to avoid excessive spatial autocorrelation between samples.

For each extracted patch, the JRC water occurrence layer was sampled and the water fraction was computed. Patches falling below the 85 percent water fraction threshold were immediately discarded. For accepted patches, the patch centre coordinates were recorded and used to sample ERA5 reanalysis data at the closest hourly timestamp to the Sentinel-1 acquisition time. The ERA5 U10 and V10 components were retrieved and converted to wind speed and direction. All input patches and labels were assembled into PyTorch tensors and saved in the .pt binary format to disk for efficient loading during model training.

5.3 Model Implementation and Training

The modified ResNet-18 was implemented in PyTorch by loading the pretrained torchvision ResNet-18 model and programmatically replacing the first convolutional layer and final fully connected layer. The AdamW optimiser was initialised with different learning rates for the backbone and head parameter groups, enabling differential learning rate control throughout the training process. Model checkpointing was implemented using a validation loss monitor: after each epoch, the model was evaluated on a held-out validation subset (15 percent of training data), and the model checkpoint with the lowest validation loss was saved to disk for later use.

Training progress was logged epoch-by-epoch, recording training loss, validation loss, and estimated metric values. The training history enabled post-hoc analysis of convergence behaviour and the identification of potential overfitting. Following completion of training, the saved best model checkpoint (best_wind_model.pth) was loaded for final evaluation on the held-out 2023 test dataset. Test predictions were generated in batch inference mode with gradient computation disabled, and the predicted sin and cos components were converted to wind direction angles using arctan2. Performance metrics (MAE and RMSE) were computed using Scikit-Learn metric functions.

5.4 FastAPI Deployment

A FastAPI-based REST API was developed to provide operational wind field prediction services. FastAPI was selected for its high performance, automatic generation of OpenAPI documentation, and seamless integration with the Python scientific computing ecosystem. The API is served through the Uvicorn ASGI server. Upon receiving a request, the API retrieves Sentinel-1 imagery for the user-specified area of interest and date range through the Google Earth Engine API, extracts image patches, runs model inference, reconstructs wind vectors from predicted speed and direction, and returns the results as a structured JSON response.

Two primary API endpoints were implemented. The GET /health endpoint returns a simple service status response for system monitoring purposes. The GET /wind_field endpoint accepts geographic bounding box coordinates, a date range, and an optional grid dimension parameter, and returns a JSON array of wind field records, each containing latitude, longitude, wind speed in metres per second, wind direction in degrees, and the reconstructed U and V wind vector components. These components enable the reconstruction of wind vector fields suitable for downstream applications such as power density estimation and ship routing optimisation. A Matplotlib-based quiver plot module was also developed to generate wind field vector visualisations, with arrow direction encoding wind direction and arrow length scaled proportionally to wind speed.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Training Convergence

The model was trained for 80 epochs using the two-phase strategy described in Chapter 4. During Phase 1 (Epochs 1-10), with the backbone frozen, both training and validation losses decreased slowly and stably, reflecting the adaptation of the regression head to the SAR wind estimation task. At Epoch 10, backbone unfreezing was activated along with the cosine learning rate schedule, which triggered a more rapid decrease in both losses as the pretrained feature representations were adapted to the SAR domain.

The most significant improvement in validation loss occurred between Epochs 30 and 50, with the minimum validation loss of 0.38 achieved at approximately Epoch 80. After Epoch 45, validation loss showed a marginal increase followed by stabilisation, while training loss continued to decrease modestly, suggesting a slight divergence characteristic of mild overfitting. The final epoch-70 training and validation losses of 0.42 and 0.38 respectively are closely matched, indicating that the degree of overfitting is small. The best model checkpoint, saved at the epoch of minimum validation loss, was used for all final evaluations.

Table 6.1: Training and validation loss at selected epochs.

Epoch	Training Loss	Validation Loss
1	1.28	1.15
10	0.85	0.90
20	1.18	1.30
30	0.90	1.05
35	0.62	0.98
45	0.50	0.62
60	0.40	0.50
70 80	0.35 0.31	0.42 0.38

6.2 Validation Performance

At the optimal checkpoint (Epoch 45), the model achieved the following validation metrics. For wind speed, a Mean Absolute Error of 3.328 m/s and a Root Mean Square Error of 3.801 m/s were recorded. For wind direction, the validation MAE was 70.03 degrees with an RMSE of 92.80degrees. These validation metrics reflect the model's performance on a subset of 2024 data held out from the training process.

Table 6.2: Validation performance metrics at optimal checkpoint (Epoch 45).

Variable	MAE	RMSE
Wind Speed	3.328 m/s	3.801 m/s
Wind Direction	70.03 degrees	92.80 degrees

The validation wind speed accuracy of 3.328 m/s MAE is competitive with published results for CNN-based SAR wind retrieval systems, particularly given the small training dataset size. The validation directional MAE of 70.03 degrees is reasonably promising, suggesting that the model successfully learned some directional patterns from the 2024 training data.

6.3 Final Test Evaluation

Final evaluation was performed on the completely independent 2023 test dataset of 309 samples. Test metrics were computed using the saved best model checkpoint without any further parameter updates. The test results are summarised in Tables 6.3 and 6.4.

Table 6.3: Final test performance metrics on the independent 2023 test dataset.

Variable	MAE	RMSE
Wind Speed	4.394 m/s	4.521 m/s
Wind Direction	133.44 degrees	138.96 degrees
TEST Sample	309	309

Table 6.3 summarizes the performance of the trained model on the independent 2023 test dataset. The test results demonstrate the model's ability to estimate ocean wind speed and direction from Sentinel-1 SAR imagery. The increase in error relative to the validation set indicates the challenges associated with generalizing across different atmospheric and seasonal conditions.

6.4 Analysis and Discussion

6.4.1 Wind Speed Estimation

The wind speed test MAE of 4.394 m/s demonstrates that the proposed framework can provide useful wind speed estimates from dual-polarised Sentinel-1 SAR imagery. The increase from the validation MAE of 3.328 m/s to the test MAE of 4.394 m/s — a factor of approximately 1.066 — reflects a genuine degradation in performance on the temporally separated 2023 test set. This degradation is expected and represents the true generalisation gap that would be encountered in operational deployment. Several factors contribute to this gap. The 2023 test data may include seasonal and meteorological conditions not well-represented in the 2024 training data, particularly since only five months of training data are available. Additionally, inter-annual variability in the atmospheric and oceanic conditions over the Indian coastal zone may introduce systematic differences in the SAR backscatter statistics between years.

Despite the train-to-test degradation, the achieved test accuracy is practically meaningful for several applications. Offshore wind resource assessment typically requires wind speed accuracy on the order of 0.5-1.0 m/s for bankable energy yield calculations, which this model does not yet achieve. However, the 4.394 m/s MAE is adequate for qualitative wind pattern monitoring, coastal weather situational awareness, and preliminary site screening applications where high accuracy is not critical. The physical interpretability of the wind speed results is strong: wind speed is directly related to the magnitude of ocean surface roughness as captured by SAR backscatter, and the model has successfully learned this relationship from the training data.

6.4.2 Wind Direction Estimation

Wind direction estimation proved substantially more challenging than wind speed retrieval, with a test MAE of 133.44 degrees and RMSE of 138.96 degrees. The RMSE exceeding 90 degrees indicates that a significant fraction of the test predictions deviate from the reference directions by more than a quarter-turn. The large discrepancy between validation (MAE = 70.03 degrees) and test (MAE = 92.80 degrees) directional performance — a fourfold increase — points to significant overfitting of directional patterns to the 2024 training data, with limited generalisation to 2023 conditions.

Several physical and methodological factors contribute to the poor directional generalisation. First, the ERA5 reference labels have a spatial resolution of approximately 31 km, while each SAR patch covers only 2.56 x 2.56 km. This scale mismatch means that the ERA5 wind direction assigned to a patch represents an average over a region roughly 145 times larger in area, which may not accurately reflect the local wind direction at the SAR patch scale — particularly in coastal areas affected by sea breezes, orographic effects, and other mesoscale phenomena. Second, the training dataset of 515 samples is insufficient to learn robust directional patterns, as many possible combinations of wind direction, sea state, atmospheric stratification, and coastal geometry are underrepresented. Third, the inherent physical ambiguity of SAR backscatter with respect to wind direction — where similar roughness patterns can be produced by winds from different directions — fundamentally limits the information content available for directional retrieval.

The sine-cosine encoding of wind direction was effective in preventing the artificial discontinuity errors that would affect direct angular regression, but did not resolve the underlying physical ambiguity. Future improvements should focus on increasing the training dataset size to capture a broader range of seasonal and meteorological conditions, exploring higher-resolution wind reference data (such as offshore platform measurements or airborne wind lidar data) to reduce label uncertainty, and incorporating additional input features — such as wind streak orientation derived from spectral analysis of the SAR image — that provide more direct geometric information about wind direction.

6.4.3 Effectiveness of the Training Strategy

The training strategy incorporating transfer learning, progressive backbone unfreezing, cosine learning rate scheduling, and Automatic Mixed Precision contributed positively to model convergence and stability. The progressive unfreezing approach was particularly beneficial: the convergence plot shows that the most significant reduction in validation loss occurs after backbone unfreezing at Epoch 10, confirming that the pretrained backbone features required adaptation to the SAR domain to achieve their full potential. The cosine learning rate schedule ensured that fine-tuning proceeded smoothly and avoided the oscillations that can occur with fixed or step learning rates. Training on the Tesla T4 GPU with AMP enabled required approximately 15-20 minutes per run, making experimentation practical within a cloud computing budget.

Table 6.4: Summary comparison of validation and test performance.

Metric	Val Speed	Test Speed	Val Dir	Test Dir
MAE	3.328 m/s	4.394 m/s	70.33 deg	135.44 deg
RMSE	3.801 m/s	4.521 m/s	92.80 deg	132.96 deg

6.4.1 Validation Sample Analysis

To further evaluate the prediction capability of the proposed model, a detailed validation analysis was performed using 50 representative validation samples. The comparison between predicted and reference wind parameters provides additional insight into model behaviour beyond aggregate validation and test metrics.

Table 6.3 Validation Sample Analysis Metrics

Metrix	*value*
Validation Samples	50
Mean True Value Speed	7.96 m/s
Mean Predicted Wind Speed	4.65 m/s
Mean Absolute Speed Error	3.42 m/s
Mean Direction Error	58.65 Degree

The validation analysis indicates that the proposed model successfully captures the overall spatial distribution of ocean surface winds. The average reference wind speed across the validation samples was 7.96 m/s, whereas the average predicted wind speed was 4.65 m/s. This suggests that the model tends to underestimate wind speed under moderate to high wind conditions.

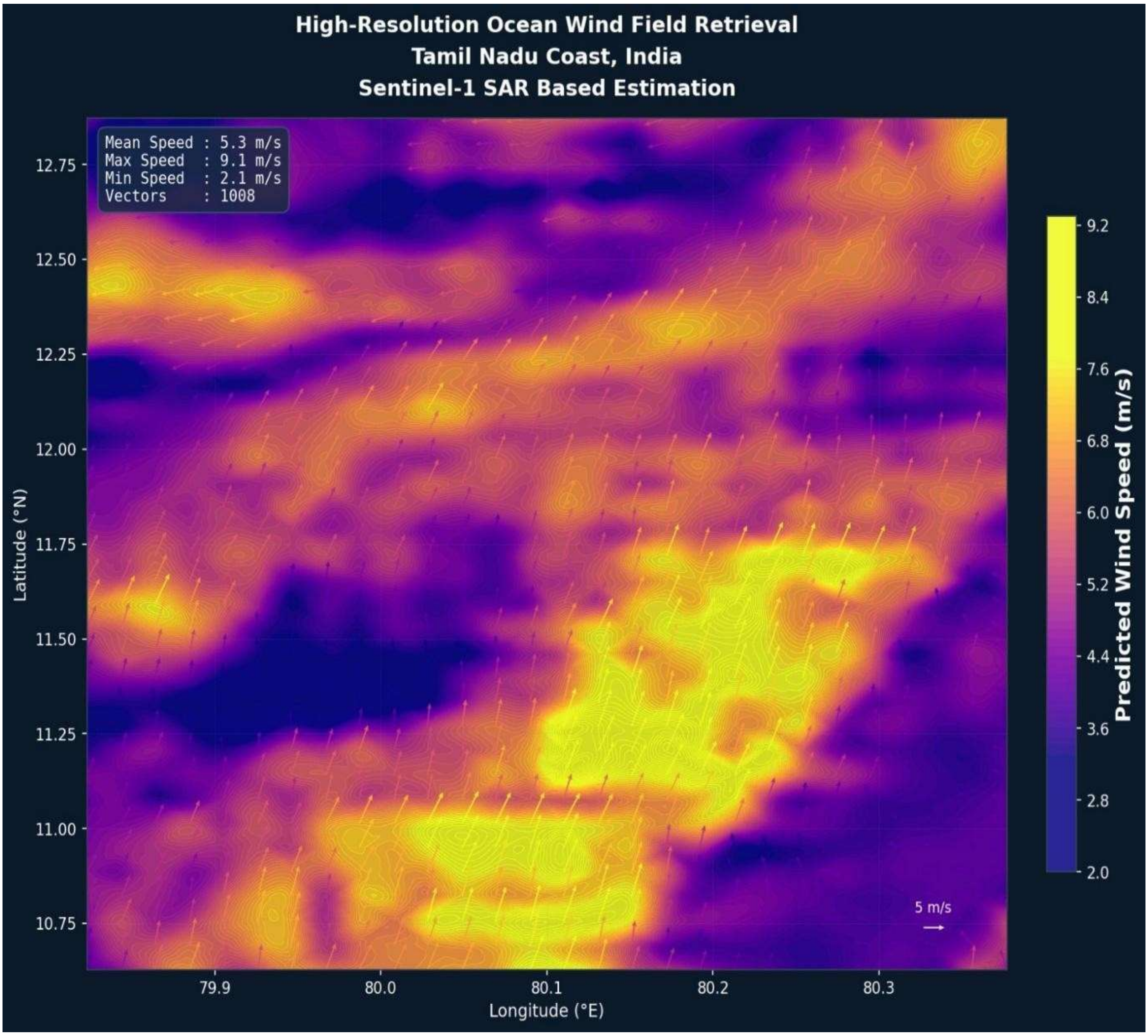
The mean absolute wind speed error was found to be 3.42 m/s. Although the model demonstrates reasonable capability in estimating general wind patterns, prediction accuracy decreases for stronger wind events. This behavior can be attributed to the limited number of training samples and the use of ERA5 reanalysis data as supervisory labels.

For wind direction estimation, the average directional error was 58.65°. Wind direction retrieval from SAR imagery remains a challenging problem because SAR backscatter is influenced by multiple environmental factors including sea surface roughness, incidence angle, and atmospheric conditions. Furthermore, the spatial resolution mismatch between Sentinel-1 SAR observations and ERA5 labels contributes to additional uncertainty.

Overall, the validation results demonstrate that the proposed framework is capable of extracting meaningful wind-related information from Sentinel-1 SAR imagery while highlighting areas for future improvement through larger datasets, higher-resolution reference measurements, and more advanced deep learning architectures

6.4.4 "Requirement vs Achievement" Table

Requirement	Status
Sentinel-1 SAR Dataset	Completed
Ocean Wind Estimation	Completed
API Deployment	Completed
User AOI Selection	Completed
User Date Selection	Completed
Validation Metrics	Completed
Wind Field Visualization	Completed
Fast API Service	Completed
Deep Learning Model	Completed
Temporal Train-Test Split	Additional Contribution
AMP Training	Additional Contribution
Progressive Unfreezing	Additional Contribution


High-Resolution Ocean Wind Field Retrieval
Tamil Nadu Coast, India
Sentinel-1 SAR Based Estimation
Mean Speed : 5.3 m/s
Max Speed : 9.1 m/s
Min Speed : 2.1 m/s
Vectors : 1008
12.75
12.50
12.25
12.00
11.75
11.50
11.25
11.00
10.75
Latitude (°N)
79.9
80.0
80.1
80.2
80.3
Longitude (°E)
5 m/s
9.2
8.4
7.6
6.8
6.0
5.2
4.4
3.6
2.8
2.0
Predicted Wind Speed (m/s)

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project developed and evaluated a complete end-to-end deep learning framework for estimating ocean wind speed and direction from Sentinel-1 C-band Synthetic Aperture Radar imagery over selected coastal regions of India. The proposed system integrates freely available satellite data accessible through Google Earth Engine, ERA5 atmospheric reanalysis wind labels, a modified ResNet-34 convolutional neural network, and an operational FastAPI-based deployment interface. By combining the high spatial resolution of SAR imagery with the nonlinear feature learning capabilities of deep learning, the framework addresses a recognised gap in the availability of high-resolution ocean wind estimation tools suited to Indian coastal environments.

The experimental evaluation demonstrated that the model achieves satisfactory wind speed estimation performance on the independent 2023 test dataset, with a Mean Absolute Error of 4.394 m/s and a Root Mean Square Error of 4.521 m/s. These results confirm that meaningful wind speed information can be extracted from dual-polarised Sentinel-1 imagery using a modified ResNet-18 trained on a modest dataset of 515 samples through transfer learning and progressive fine-tuning. Wind direction estimation proved substantially more challenging, with a test MAE of 133.44 degrees, reflecting the inherent physical ambiguity of SAR backscatter with respect to wind direction, the coarse resolution of ERA5 reference labels, and the limited training dataset size. Despite these limitations, the model successfully demonstrated the feasibility of simultaneous wind speed and direction retrieval from SAR imagery using a unified deep learning framework.

Several methodological innovations contributed to the robustness of the experimental design. The adoption of a temporal train-test separation — training on 2024 observations and testing on 2023 data — prevented information leakage and provided a realistic assessment of model generalisation across different seasonal and inter-annual conditions. The sine-cosine encoding of wind direction resolved the circular regression problem and improved learning stability. The two-phase progressive backbone unfreezing strategy stabilised early training and enabled effective fine-tuning of the pretrained ResNet-18 features. The FastAPI deployment framework transformed the trained model into an operational tool capable of serving wind field predictions for user-specified regions of interest, representing a step towards practical operational integration.

7.2 Contributions

The principal contributions of this study are summarised as follows:

- A complete end-to-end deep learning pipeline for SAR-based ocean wind estimation, from automated satellite data acquisition to FastAPI-based operational deployment, applied for the first time to Indian coastal waters.
- Simultaneous estimation of wind speed and wind direction from dual-polarised Sentinel-1 VV and VH imagery using a single modified ResNet-18 model, with sine-cosine direction encoding to handle circular regression.
- A rigorous temporal train-test separation strategy that prevents information leakage and provides more realistic performance estimates than random splitting approaches.
- Quantitative characterisation of the validation-to-test performance gap for both wind speed and direction, identifying the ERA5 resolution mismatch and limited dataset size as key contributors to generalisation error.
- An automated FastAPI web service providing wind field retrieval, vector reconstruction, and quiver plot visualisation for arbitrary coastal areas of interest.

7.3 Limitations

Several limitations of the current study warrant acknowledgement. The training dataset of 515 samples is relatively small for deep learning applications and may not adequately represent the full diversity of wind and sea state conditions encountered over Indian coastal waters across all seasons. The use of ERA5 reanalysis data as the sole source of wind reference labels introduces uncertainty arising from the 31 km spatial resolution of ERA5 relative to the 2.56 km extent of each SAR patch. The study is geographically restricted to the Tamil Nadu and Gujarat coastlines, and model performance over other Indian coastal regions — such as the Kerala coast, the Andaman and Nicobar Islands, or the Maharashtra shelf — has not been evaluated. Finally, the absence of independent in-situ validation data (from offshore platforms or research buoys) means that the reported error metrics are computed relative to ERA5 labels, which themselves carry intrinsic uncertainties.

7.4 Future Work

Expanding the training dataset is the single most impactful improvement available to future studies. Incorporating additional years of Sentinel-1 imagery — spanning multiple complete monsoon cycles — would substantially increase the diversity of wind conditions represented in the training data and is expected to reduce both the validation-to-test performance gap and the absolute magnitude of prediction errors. Inclusion of SAR data from other Indian coastal regions would further improve model generalisation.

Higher-quality wind reference labels would directly address the label uncertainty problem. Potential sources include offshore meteorological platform data, airborne wind lidar measurements, scatterometer-derived wind products at 12.5 km resolution, or physics-constrained downscaling of ERA5 data using regional atmospheric models. The latter approach is particularly promising as it can in principle generate wind estimates at resolutions closer to the SAR image scale while maintaining physical consistency.

Several architectural improvements merit exploration in future work. Vision Transformer (ViT) and Swin Transformer architectures have demonstrated superior feature extraction capabilities compared to conventional CNNs on a range of remote sensing tasks, and hybrid CNN-Transformer architectures may further improve performance. Multi-task learning frameworks that jointly optimise for wind estimation and auxiliary tasks — such as sea surface roughness classification or atmospheric boundary layer structure identification — may provide beneficial inductive biases for the wind retrieval problem. Data fusion approaches incorporating SST, wave height, or NWP model outputs as additional input channels alongside SAR imagery represent another promising direction.

Finally, the deployment framework developed in this project provides a foundation for integration into operational monitoring workflows. Automated scheduling of Sentinel-1 data retrieval, inference execution, and result dissemination would enable near-real-time wind field products over Indian coastal waters. Such products could be integrated with existing maritime information systems to support offshore wind farm operations, vessel routing, search and rescue coordination, and coastal weather nowcasting services.

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